LOGARITHMS ON ONE PAGE

$\log_b x$ undoes b^x

A logarithm answers one question: which power of the base gives this number?

THE DEFINITION

A logarithm is an exponent

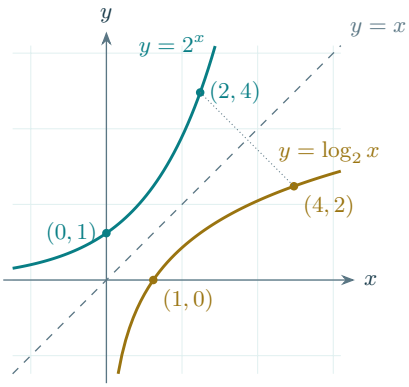
For a base $b > 0$ with $b \neq 1$ and a number $x > 0$,

$$y \;=\; \log_b x \qquad \Longleftrightarrow \qquad b^{\,y} \;=\; x$$

so $\log_2 8 = 3$ because $2^3 = 8$, $\log_{10} 0.01 = -2$ because $10^{-2} = 0.01$, and $\log_9 3 = \frac{1}{2}$ because $\sqrt{9} = 3$. Applying one function after the other gets you back where you started:

$$b^{\log_b x} = x, \qquad \log_b(b^x) = x, \qquad \log_b 1 = 0, \qquad \log_b b = 1.$$

The two special bases have their own names: $\ln x = \log_{\mathrm{e}} x$ is the natural logarithm and $\lg x = \log_{10} x$ is the common one.



Swapping input and output reflects each graph onto the other in $y = x$.

THREE LAWS, ONE SOURCE

Every log law is an index law read backwards

Writing $x = b^m$ and $y = b^n$, so that $m = \log_b x$ and $n = \log_b y$, each index law turns into a log law by taking $\log_b$ of both sides.

$b^m\, b^n \;=\; b^{m+n}$
$$\log_b(xy) \;=\; \log_b x + \log_b y$$

$b^m / b^n \;=\; b^{m-n}$
$$\log_b\left(\frac{x}{y}\right) \;=\; \log_b x - \log_b y$$

$(b^m)^k \;=\; b^{mk}$
$$\log_b\left(x^k\right) \;=\; k \log_b x$$

Watch for: there is no law for $\log_b(x + y)$; the expression $(\log_b x)^2$ is a different thing from $\log_b x^2 = 2\log_b x$; and $\dfrac{\ln x}{\ln y}$ is a different thing from $\ln \dfrac{x}{y}$.

CHANGE OF BASE

Any log from $\ln$

Taking $\ln$ of $b^{\,y} = x$ gives $y \ln b = \ln x$, so

$$\log_b x \;=\; \frac{\ln x}{\ln b} \;=\; \frac{\log_a x}{\log_a b}$$

for any convenient base a. For example $\log_2 10 = \dfrac{\ln 10}{\ln 2} \approx 3.32$, which says that ten lies a little past 2^3.

SOLVING

Bring the unknown down from the exponent

Isolate the power, then take logs:

$$3 \cdot 2^x = 50 \;\Rightarrow\; 2^x = \tfrac{50}{3} \;\Rightarrow\; x = \frac{\ln(50/3)}{\ln 2} \approx 4.06.$$

Combining logs can create false roots, so check the domain at the end:

$$\log_2 x + \log_2(x-2) = 3 \;\Rightarrow\; x(x-2) = 8 \;\Rightarrow\; x = 4 \text{ or } x = -2,$$

and only $x = 4$ survives, since $\log_2(-2)$ is undefined.

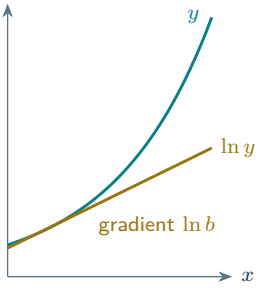
LOGS STRAIGHTEN CURVES

Turning exponentials and powers into straight lines

Taking $\ln$ of a model turns multiplication into addition, which turns a curve into a line whose gradient and intercept you can read off:

$$y = A\,b^x \;\Rightarrow\; \ln y = \ln A + x \ln b, \qquad y = A\,x^n \;\Rightarrow\; \ln y = \ln A + n \ln x.$$

Plot $\ln y$ against x for growth and decay, where the gradient is $\ln b$, and $\ln y$ against $\ln x$ for a power law, where the gradient is the power n; in both cases the intercept is $\ln A$.



Conventions. On most calculators the `log` key means $\log_{10}$ and the `ln` key means $\log_{\mathrm{e}}$. Only positive numbers have real logarithms, and the base must be positive and different from 1, because no power of 1 can give anything except 1.